

# Abstract Algebra - Stockholm University 2025

## 1. Groups

### 1.1. Conjugation, centralizer, center, normalizer

For  $g \in G$ , the conjugation map  $c_g(x) = gxg^{-1}$  is an automorphism of  $G$  (an *inner* automorphism). For a subset  $S \subseteq G$ , the centralizer is

$$C_G(S) = \{ g \in G : gs = sg \text{ for all } s \in S \}; \quad C_G(g) := C_G(\{g\}).$$

The center is

$$Z(G) = C_G(G) = \{ g \in G : gx = xg \ \forall x \in G \}$$

(always a characteristic, hence normal, subgroup). For a subgroup  $H \leq G$ , the normalizer is

$$N_G(H) = \{ g \in G : gHg^{-1} = H \},$$

the largest subgroup of  $G$  in which  $H$  is normal (so  $H \trianglelefteq N_G(H)$ ). One always has  $C_G(H) \leq N_G(H)$ . The number of conjugates of  $H$  in  $G$  equals  $[G : N_G(H)]$ .

### 1.2. Normal subgroups and quotients

A subgroup  $N \leq G$  is normal (write  $N \trianglelefteq G$ ) if  $gNg^{-1} = N$  for all  $g \in G$ . Equivalently, every left coset equals the corresponding right coset:  $gN = Ng$ . If  $N \trianglelefteq G$ , the set of cosets  $G/N = \{gN : g \in G\}$  is a group with  $(gN)(hN) = (gh)N$ .

### 1.3. Cosets and index

For  $H \leq G$  and  $g \in G$ , the left (resp. right) coset is  $gH = \{gh : h \in H\}$  (resp.  $Hg = \{hg : h \in H\}$ ). Left cosets are either disjoint or equal and they partition  $G$ . Each left coset  $gH$  has  $|H|$  elements via the bijection  $H \rightarrow gH, h \mapsto gh$ . The *index* of  $H$  in  $G$  is the number of left cosets, denoted  $[G:H]$ .

### 1.4. Lagrange's Theorem

If  $G$  is finite and  $H \leq G$ , then

$$|G| = [G:H] \cdot |H|.$$

*Proof sketch.*  $G$  is the disjoint union of  $[G:H]$  left cosets, each of cardinality  $|H|$ .

## Consequences

- $|H| \mid |G|$  for every  $H \leq G$ .
- For  $g \in G$ , the order  $|g|$  divides  $|G|$ ; hence  $g^{|G|} = e$ .
- Index multiplicativity: for  $K \leq H \leq G$  (with  $G$  finite),  $[G:K] = [G:H][H:K]$ .
- If  $[G:H] = 2$ , then  $H \trianglelefteq G$  (there are only two cosets, so  $gHg^{-1}$  must be  $H$ ).

**Groups of prime order are cyclic.** If  $|G| = p$  (prime) and  $g \in G$ ,  $g \neq e$ , then  $|\langle g \rangle| = |g| \mid p$  by Lagrange, so  $|g| = p$  and  $\langle g \rangle = G$ . Hence  $G$  is cyclic and  $G \cong \mathbb{Z}/p\mathbb{Z}$ .

## 1.5. Isomorphism theorems

- (1) **First isomorphism theorem.** For  $\varphi : G \rightarrow H$  a homomorphism,

$$G/\ker \varphi \cong \operatorname{im} \varphi.$$

- (2) **Second isomorphism theorem.** For  $A \leq G$  and  $N \trianglelefteq G$ ,

$$A \cap N \trianglelefteq A, \quad AN \leq G, \quad AN/N \cong A/(A \cap N).$$

- (3) **Third isomorphism theorem.** If  $K \leq H \trianglelefteq G$ , then  $H/K \trianglelefteq G/K$  and

$$(G/K)/(H/K) \cong G/H.$$

- (4) **Correspondence (lattice) theorem.** For  $N \trianglelefteq G$ , the map

$$\{\text{subgroups } H \mid N \leq H \leq G\} \longleftrightarrow \{\text{subgroups of } G/N\}, \quad H \mapsto H/N$$

is a bijection preserving inclusion and index; it carries normal subgroups  $H \trianglelefteq G$  with  $N \leq H$  to normal subgroups of  $G/N$ .

## 2. Group actions

### 2.1. Actions on cosets and by conjugation

- For  $H \leq G$ , the action  $G \curvearrowright G/H$  by left multiplication is transitive, with stabilizer of  $aH$  equal to  $aHa^{-1}$ .
- The action  $G \curvearrowright G$  by conjugation,  $g \cdot x = gxg^{-1}$ , partitions  $G$  into conjugacy classes. The stabilizer of  $x$  is its centralizer  $C_G(x)$ , and  $|\operatorname{Orb}(x)| = [G : C_G(x)]$ .

### 2.2. Cayley's Theorem

Every group  $G$  is isomorphic to a subgroup  $H$  of the symmetric group  $S_{|G|}$ :

$$G \cong H \leq S_{|G|}.$$

### 2.3. Orbit–stabilizer

For any action  $G \curvearrowright X$  and any  $x \in X$  there is a canonical bijection

$$G/G_x \xrightarrow{\sim} \text{Orb}(x), \quad gG_x \mapsto g \cdot x.$$

In particular, if  $G$  is finite then

$$|\text{Orb}(x)| = [G : G_x] \quad \text{and} \quad |G| = |\text{Orb}(x)| \cdot |G_x|.$$

### 2.4. Class Equation

For a finite group  $G$ , let  $Z(G)$  be the center of  $G$ . Then

$$|G| = |Z(G)| + \sum_{i=1}^r [G : C_G(g_i)],$$

where  $g_1, \dots, g_r$  are representatives of the conjugacy classes not contained in  $Z(G)$ .

### 2.5. Conjugacy in $S_n$

Two permutations in  $S_n$  are conjugate if and only if they have the same cycle type (i.e. the same cycle lengths with the same multiplicities).

$$\sigma, \tau \in S_n \quad \text{are conjugate} \iff \text{cycle-type}(\sigma) = \text{cycle-type}(\tau).$$

Hence, the conjugacy classes of  $S_n$  correspond bijectively to partitions of  $n$ .

### 2.6. Automorphisms of cyclic groups

For  $n \geq 1$ , the automorphism group of the cyclic group  $C_n \cong \mathbb{Z}/n\mathbb{Z}$  is

$$\text{Aut}(C_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times,$$

hence  $|\text{Aut}(C_n)| = \varphi(n)$ . Here  $\varphi(n)$  is *Euler's totient function*:

$$\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times| = |\{1 \leq k \leq n : \gcd(k, n) = 1\}|.$$

Equivalently, if  $n = \prod p_i^{e_i}$  then

$$\varphi(n) = n \prod_i \left(1 - \frac{1}{p_i}\right) = \prod_i (p_i^{e_i} - p_i^{e_i-1}).$$

## 3. Sylows Theorem

Let  $|G| = p^a m$  with  $p \nmid m$  and  $a \geq 1$ . A subgroup of order  $p^a$  is a *Sylow  $p$ -subgroup*. Write  $\text{Syl}_p(G)$  for the set of Sylow  $p$ -subgroups and  $n_p := |\text{Syl}_p(G)|$ .

### 3.1. Statements

- $\text{Syl}_p(G) \neq \emptyset$ .
- Every  $p$ -subgroup of  $G$  is contained in some conjugate of a Sylow  $p$ -subgroup. In particular, any two Sylow  $p$ -subgroups of  $G$  are conjugate in  $G$ .
- $G$  acts on  $\text{Syl}_p(G)$  by  $g \cdot P = gPg^{-1}$ . For  $P \in \text{Syl}_p(G)$ , the stabilizer is  $N_G(P)$ , hence

$$n_p = [G : N_G(P)].$$

- $n_p \equiv 1 \pmod{p}$  and  $n_p \mid m$ .
- If  $n_p = 1$ , then the unique Sylow  $p$ -subgroup  $P$  is normal *and characteristic* in  $G$  (automorphisms permute Sylow  $p$ -subgroups; uniqueness forces  $P$  to be fixed).

### 3.2. Fundamental Theorem of Finitely Generated Abelian Groups

Every finitely generated abelian group  $G$  decomposes uniquely (up to isomorphism) as

$$G \cong \mathbb{Z}^r \times \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_s}, \quad \text{with } r \geq 0, n_j \geq 2, \text{ and } n_{i+1} \mid n_i.$$

**Consequences.**  $G \cong \mathbb{Z}^r \times T(G)$  where  $T(G)$  is the (finite) torsion subgroup. If  $|G| = \prod p_i^{a_i}$  then the number of abelian groups of order  $|G|$  (up to isomorphism) is

$$\prod_i p(a_i),$$

where  $p(k)$  is the number of partitions of  $k$ .

## 4. Rings

### 4.1. Ideals and basic ideal arithmetic

Throughout,  $R$  is a commutative ring with 1.

**Ideal.** An *ideal*  $I \trianglelefteq R$  is an additive subgroup  $I \leq (R, +)$  with the absorption property:  $rI \subseteq I$  for all  $r \in R$  (equivalently  $Ir \subseteq I$  in the commutative case).

**Principal ideal.** For  $a \in R$ ,  $(a) := Ra = \{ra : r \in R\}$ .

**Zero and unit ideals.**  $0 := \{0\}$  and  $R$  are ideals. Every ideal contains 0.  $1 \in I$  iff  $I = R$  (the *unit ideal*); otherwise  $1 \notin I$ .

**Sum and intersection.** For ideals  $I, J \trianglelefteq R$ ,

$$I + J := \{i + j : i \in I, j \in J\}, \quad I \cap J \trianglelefteq R.$$

$I + J$  is the *smallest* ideal containing  $I \cup J$ ;  $I \cap J$  is the *largest* ideal contained in both.

**Product.**

$$IJ := \left\{ \sum_{k=1}^m i_k j_k : i_k \in I, j_k \in J, m \geq 1 \right\}.$$

Always  $IJ \subseteq I \cap J$ . If  $I + J = R$  (i.e.  $I, J$  are *comaximal*), then  $IJ = I \cap J$ .

**Powers.**  $I^n := \underbrace{I \cdot I \cdots I}_{n \text{ times}}$ ; then  $I^{n+1} \subseteq I^n$ .

**Quotients.**  $R/I$  is a ring; the projection  $\pi : R \rightarrow R/I$  is surjective with  $\ker \pi = I$ . Ideals  $J$  with  $I \subseteq J \leq R$  correspond to ideals of  $R/I$  via  $J \mapsto J/I$ .

## 4.2. Chinese Remainder Theorem

Let  $R$  be a commutative ring with 1 and let  $A_1, \dots, A_k \leq R$  be ideals. If the  $A_i$  are pairwise comaximal ( $A_i + A_j = R$  for  $i \neq j$ ), then the natural map

$$\phi : R \longrightarrow \prod_{i=1}^k R/A_i, \quad r \mapsto (r + A_i)_i$$

is surjective with kernel  $\bigcap_{i=1}^k A_i = \prod_{i=1}^k A_i$ . Hence

$$R / \left( \prod_{i=1}^k A_i \right) \cong \prod_{i=1}^k R/A_i.$$

## 5. Integral -, Unique Factorization - & Principle Ideal Domains

### 5.1. Definitions

**Integral domain.** A (commutative, unital) ring  $R$  with  $1 \neq 0$  and no zero divisors.

**Unique Factorization Domain (UFD).** An integral domain in which every nonzero nonunit factors as a finite product of irreducibles, and this factorization is unique up to order and associates.

**Principal Ideal Domain (PID).** An integral domain in which every ideal is principal, i.e.  $\forall I \leq R$  there exists  $a \in R$  with  $I = (a)$ .

**Euclidean domain (ED).** An integral domain  $R$  with a function  $\delta : R \setminus \{0\} \rightarrow \mathbb{N}$  such that for all  $a, b \in R$  with  $b \neq 0$  there exist  $q, r \in R$  with  $a = bq + r$  and either  $r = 0$  or  $\delta(r) < \delta(b)$  (a “division algorithm”).

**Fields are Euclidean.** For a field  $F$ , define  $\delta(x) = 0$  for  $x \neq 0$ . Then for any  $a, b \in F$  with  $b \neq 0$ ,  $a = b(ab^{-1}) + 0$ , so  $F$  is Euclidean (hence a PID and a UFD).

### Implications

$$\text{Field} \Rightarrow \text{Euclidean domain} \Rightarrow \text{PID} \Rightarrow \text{UFD} \Rightarrow \text{integral domain}.$$

## 5.2. Gauss's Lemma

Let  $R$  be a UFD with field of fractions  $F$ , and let  $p(x) \in R[x]$ . If  $p$  is reducible in  $F[x]$ , then  $p$  is reducible in  $R[x]$ . (Equivalently: if  $p$  is irreducible in  $R[x]$ , then it is irreducible in  $F[x]$ .)

## 5.3. Eisenstein's Criterion

Let  $R$  be an integral domain and  $P \subset R$  a prime ideal. Consider

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 \in R[x].$$

If  $a_{n-1}, \dots, a_0 \in P$  and  $a_0 \notin P^2$ , then  $f(x)$  is irreducible in  $R[x]$ .

**Corollary ( $\mathbb{Z}[x]$ ).** Let  $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$ . If there exists a prime  $p$  such that  $p \mid a_i$  for  $i = 0, \dots, n-1$  and  $p^2 \nmid a_0$ , then  $f$  is irreducible in  $\mathbb{Z}[x]$  (hence in  $\mathbb{Q}[x]$  by Gauss).

## 6. Field Extensions

### 6.1. Simple extensions and minimal polynomials

If  $\alpha$  is algebraic over a field  $F$ , there is a unique monic irreducible  $m_\alpha(x) \in F[x]$  with  $m_\alpha(\alpha) = 0$  (the *minimal polynomial* of  $\alpha$ ).

### 6.2. Construction via quotients

If  $p(x) \in F[x]$  is irreducible and  $K/F$  contains a root  $\alpha$  of  $p$ , then

$$F(\alpha) \cong F[x]/(p(x)).$$

In particular, for any irreducible  $p$ , the quotient  $F[x]/(p)$  is a field of degree  $\deg p$ , and the class  $\bar{x} := x + (p)$  is a root of  $p$ .

### 6.3. Degree and basis of a simple extension

Let  $\alpha$  be algebraic over  $F$  with  $m_\alpha$  of degree  $n$ . Then

$$[F(\alpha) : F] = n, \quad \{1, \alpha, \alpha^2, \dots, \alpha^{n-1}\} \text{ is an } F\text{-basis of } F(\alpha).$$

### 6.4. Tower law

For fields  $F \subseteq K \subseteq L$ ,

$$[L : F] = [L : K][K : F].$$

### 6.5. Finite $\Rightarrow$ algebraic

If  $K/F$  is a finite extension, then every  $\alpha \in K$  is algebraic over  $F$  (hence  $K/F$  is algebraic).